

Five agents, one platform

— *each with one job, all auditable.*

Origin runs a small, specialised agent stack rather than one monolithic AI. A cheap router decides which specialist hears each request; only the agents that need expensive reasoning pay for it. Every call is hash-chained for replay. Every output that matters to the regulator passes a 3-person sign-off chain before it's recorded as fact.

01 The 5 agents

Each has one job, one model size matched to it, one place it shows up.

Maestro

Haiku 4.5

ROUTER · CHEAP + FAST

Single job: classify each chat message into one of 6 intents and dispatch to the right specialist. ~1–2 second classification. Returns JSON. Costs less than R0.01 per call.

Where: ChatFab → POST /api/chat SLA: <2s

Owner Concierge

Opus 4.7 · adaptive

CROSS-UNIT SYNTHESIS

On demand: bundles cannabis state + ticket queues + SMF status + sign-off bottlenecks and produces a 200–350-word morning brief naming actual staff and citing real numbers.

Where: Owner Dashboard widget SLA: ~15s

SMF Composer

Opus 4.7 · adaptive

REGULATOR-GRADE PROSE

On demand: takes a stale SMF section + fresh facility data, drafts a revised body in regulator-grade GMP prose. Output never auto-applies — RP reviews, then signs the chain.

Where: SMF section modal SLA: ~14s

General Ops Q&A

Sonnet 4.6

FREE-FORM CHAT ANSWERS

Handles "what's overdue?", "which sections are stale?", "how is the cultivation team today?". Sees a live state digest; system-prompt-bound to never invent facts.

Where: ChatFab default SLA: ~5–8s

SMF Drift Daemon

deterministic

ALWAYS-ON EVENT LISTENER

No LLM. Pure rules. Subscribes to 16 event types. When an event invalidates a signed SMF section (new greenhouse, SOP revision, calibration drift, etc.), it marks the section stale and raises a ticket for the RP.

Where: background process SLA: <200ms

Audit hooks

infrastructure

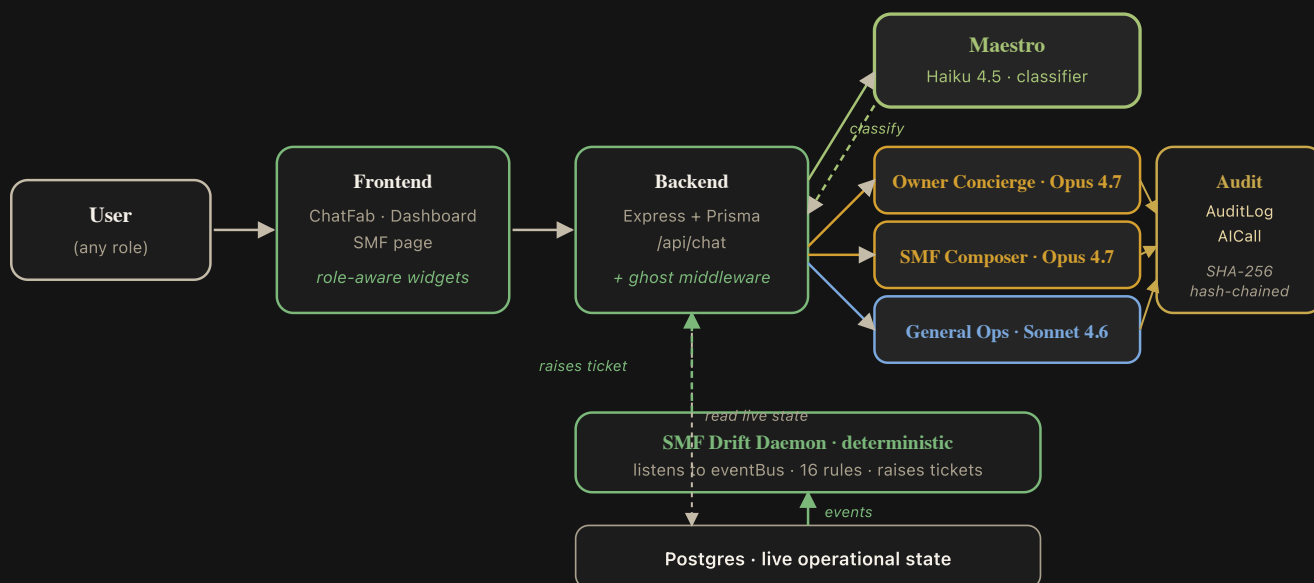
HASH-CHAINED LOG OF EVERY CALL

Two append-only tables, both SHA-256 chained: **AuditLog** for human + system actions, **AICall** for every Claude call (prompt, response, sources, tokens, real actor + ghost target).

Where: Postgres tables SLA: instant

02 System diagram

How a chat message flows from user through Maestro to a specialist and back, with audit on the side.



Flow A · A user asks a question

Renae types "What's overdue right now?" into the floating chat. End-to-end, ~7 seconds.

1

USER · RENAE

Types message into ChatFab (floating bubble bottom-right of every dashboard). `POST /api/chat` with `{message: "What's overdue right now?"}`.

2

BACKEND · AUTH + GHOST MIDDLEWARE

JWT verified. Ghost-mode header checked (none here). Request context set with `realUserId = renae`. Wrapped in `AsyncLocalStorage` for downstream services.

3

MAESTRO · HAIKU 4.5

~1.5s. Classifies intent: `{intent: "TICKETS_QUERY", reason: "User is asking about overdue tickets and assignment status."}`. Logs the call to `AICall`.

4

BACKEND · DISPATCH

Switch on intent. `TICKETS_QUERY` → routes to General Ops with the user's role + name + tenantId.

5

GENERAL OPS · SONNET 4.6

Pulls live state digest from Postgres (cultivation + tickets + SMF + key personnel). System prompt forbids invention. ~5s. Returns markdown answer + sources object. Logs to `AICall`.

6

FRONTEND · CHATFAB

Renders the markdown answer with route badge ("Tickets Q&A") + sources footer ("50 plants · 1 open ticket · 2 approved sections · 56 drafts · ..."). Renae can read the answer AND see the data that grounded it.

04 Flow B · The drift loop (the regulator wow)

Operational change → automatic stale flag → AI redraft → 3-stage human sign-off → frozen snapshot.

1

USER · RENAE (FACILITY MANAGER)

Adds a new greenhouse via BayGrid. Backend emits `GREENHOUSE_CREATED` event onto eventBus.

2

SMF DRIFT DAEMON · DETERMINISTIC

Listening on eventBus. Rule says `GREENHOUSE_CREATED` affects sections C.3.1, C.1.6, C.1.6.2. Finds C.3.1 is currently `AR_APPROVED` (signed) — marks it `staleSince = now` with the reason. Raises an `SMF_DRIFT` ticket assigned to `RESPONSIBLE_PHARMACIST`.

3

USER · BERNE (RP)

Sees ticket in her queue. Opens SMF page → C.3.1 has a stale amber banner. Clicks **"Draft with AI"**.

4

SMF COMPOSER · OPUS 4.7 + ADAPTIVE

~14s. Pulls fresh facility data (greenhouses, rooms, equipment, water spec). Drafts revised body in GMP prose. Mentions the new greenhouse by name. Ends with *"Outstanding:"* line listing what RP still needs to gather. Logs everything to `AICall`.

5

FRONTEND · SMF SECTION MODAL

Renders the draft below the current body in a **RED-bordered** card with **"AI-GENERATED · NOT YET APPROVED"** banner. Apply / Discard buttons.

6

USER · BERNE REVIEWS + APPLIES

Reads draft. Confirms accuracy. Clicks **"Apply"**. The draft text is copied into `bodyText`. **Sign-off chain resets to DRAFT** — all 3 signatures cleared.

7

SIGN-OFF CHAIN · RP → DAR → AR

Berne clicks **"RP Sign"** → status `RP_SIGNED`. Coenie logs in, clicks **"DAR Sign"** → `DAR_SIGNED`. Ilse logs in, clicks **"AR Approve & Freeze v"** → `AR_APPROVED`. New `SMFSectionVersion` snapshot frozen with all 3 signatory IDs + timestamps.

8

SYSTEM · AUDIT CLOSED

Drift ticket auto-resolves on next poll. AuditLog captures every step. AICall captures the model calls. Both hash-chained. SAHPRA can later request "show me how C.3.1 v2 was written" and we replay the entire trail.

05 **Capability matrix**

Which agent serves which role, when, and at what cost.

AGENT	MODEL	USED BY	TRIGGERED WHEN	PER CALL
Maestro	Haiku 4.5	Every authed user (chat)	Every chat message	~R0.005
Owner Concierge	Opus 4.7	Owner trio (Ilse · Coenie · Flo)	Owner clicks "Generate brief" or chats "morning brief"	~R8
SMF Composer	Opus 4.7	RP + Owner-tier (level 4+)	"Draft with AI" on a stale section	~R6
General Ops	Sonnet 4.6	Every authed user (chat)	Maestro routes to it	~R0.50
SMF Drift Daemon	deterministic	Background — affects RP's queue	16 event types	R0
Audit hooks	deterministic	Inspector + ops	Every action by every actor	R0

Total expected runtime: ~R1,300–R1,700/month across all agents at projected staff usage. Engineering time, not token cost, was always the constraint.

06 **What's deliberately not in the stack yet**

Honest scoping — these slot in next, not tonight.

AGENT	WHY DEFERRED	WHEN IT LANDS
Chicken Biosec Daemon (Sonnet 4.6)	Waits for chicken module schema + flock data	Phase 2 — alongside Poultry Master File
SAHPRA Section 21 packet generator	Patient-side workflow not active yet	When Origin Retail handoff lands
RP Auditor agent	Only meaningful after S21 packet generator ships	After SAHPRA S21 work
Tool-use loop (PTC)	Current agents need only digest data	If chat needs cross-system actions